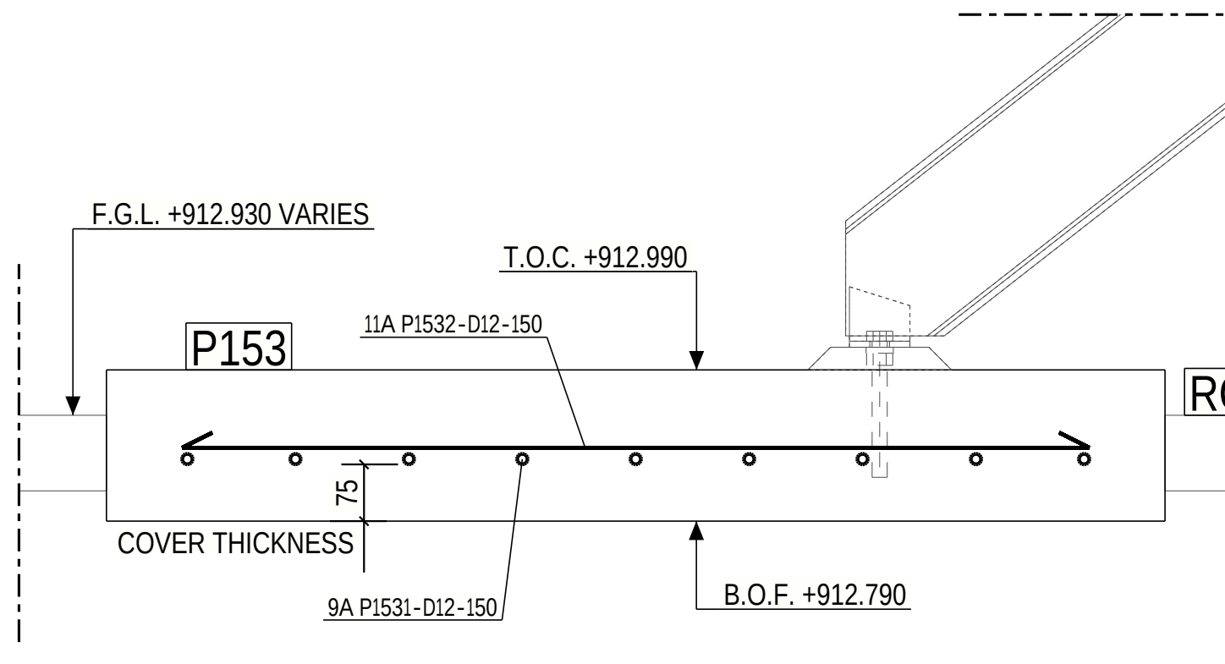
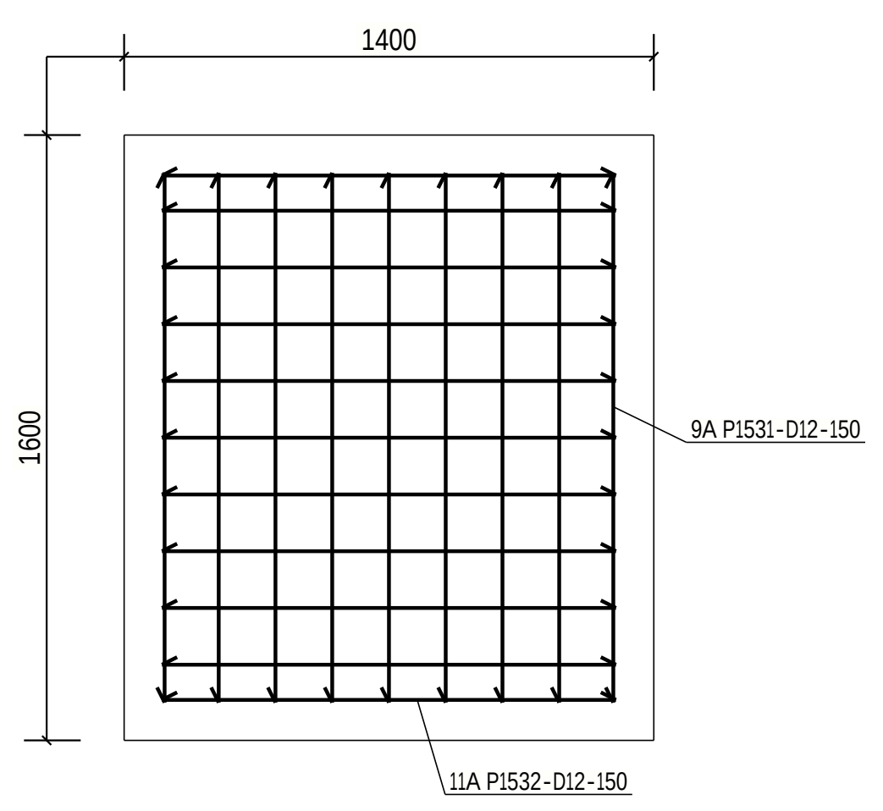
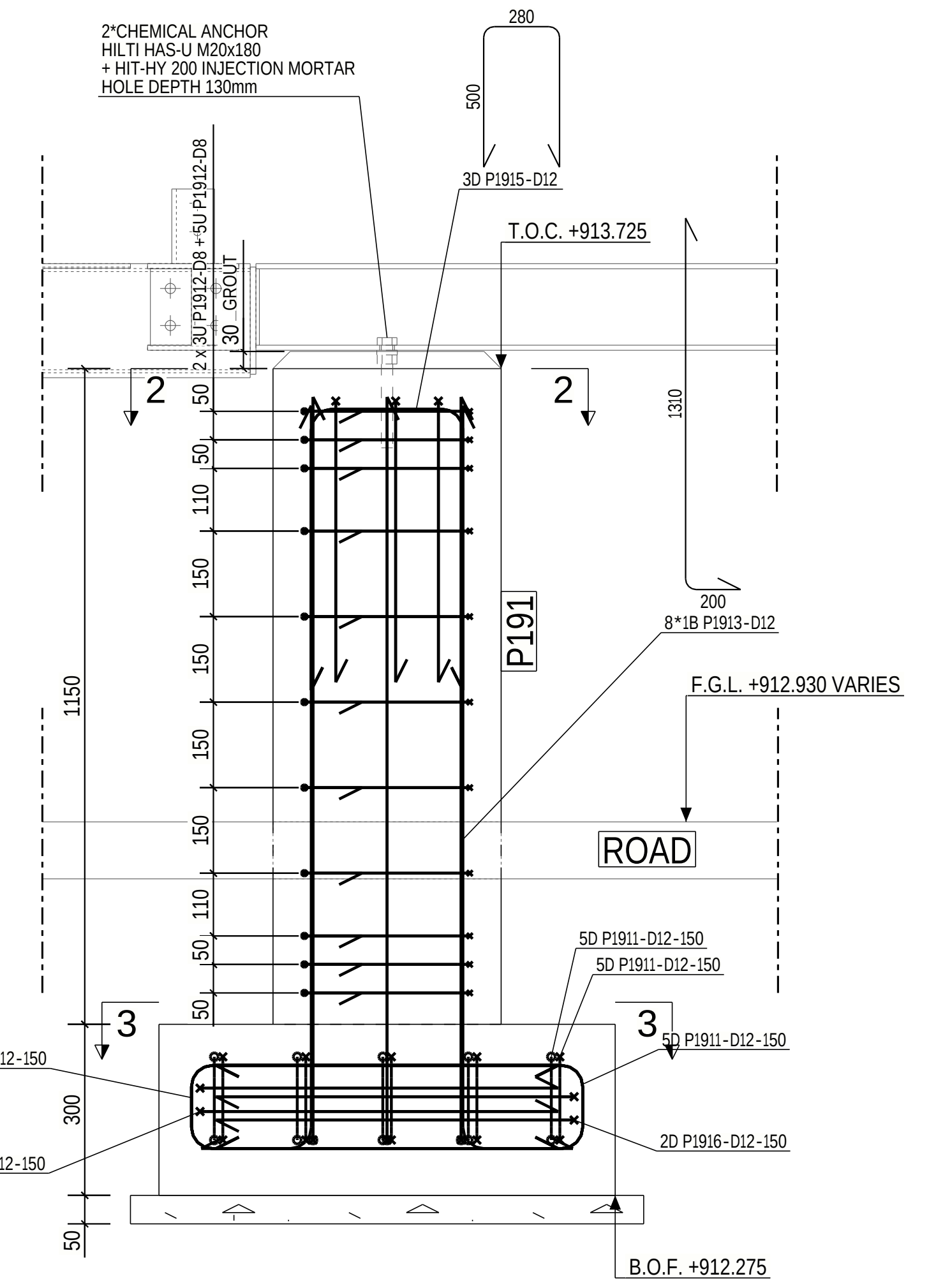
MD-216-49W1-EG-CV-DED-1038
DRAWING NUMBER
D01
REV


F.G.L. +912.930 VARIES
T.O.C. +912.990
11A P1532-D12-150
P153
ROAD
1
1
COVER THICKNESS
75
9A P1531-D12-150
B.O.F. +912.790

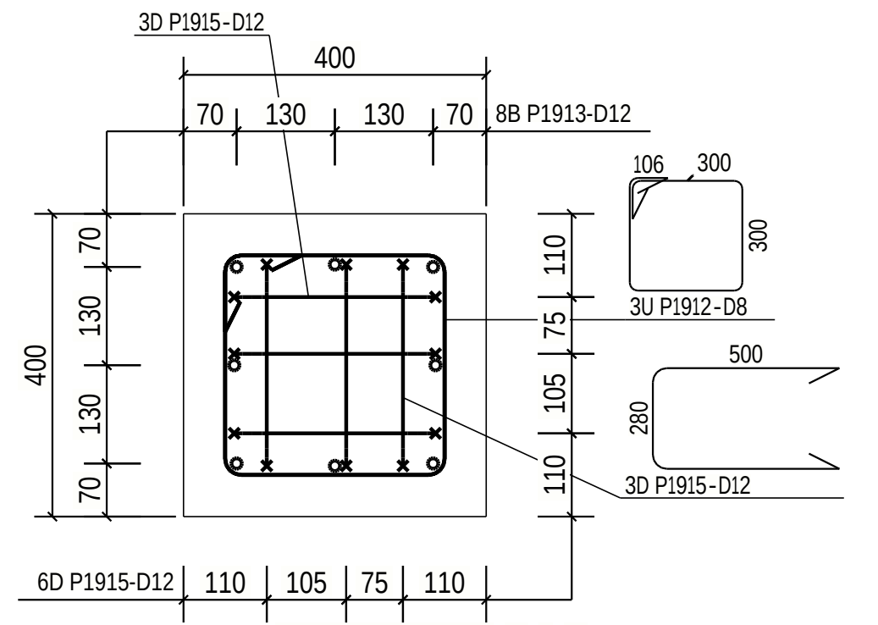
STAIR LANDING P153 REINFORCEMENT
(SCALE 1:10)


1400
1600
9A P1531-D12-150
11A P1532-D12-150

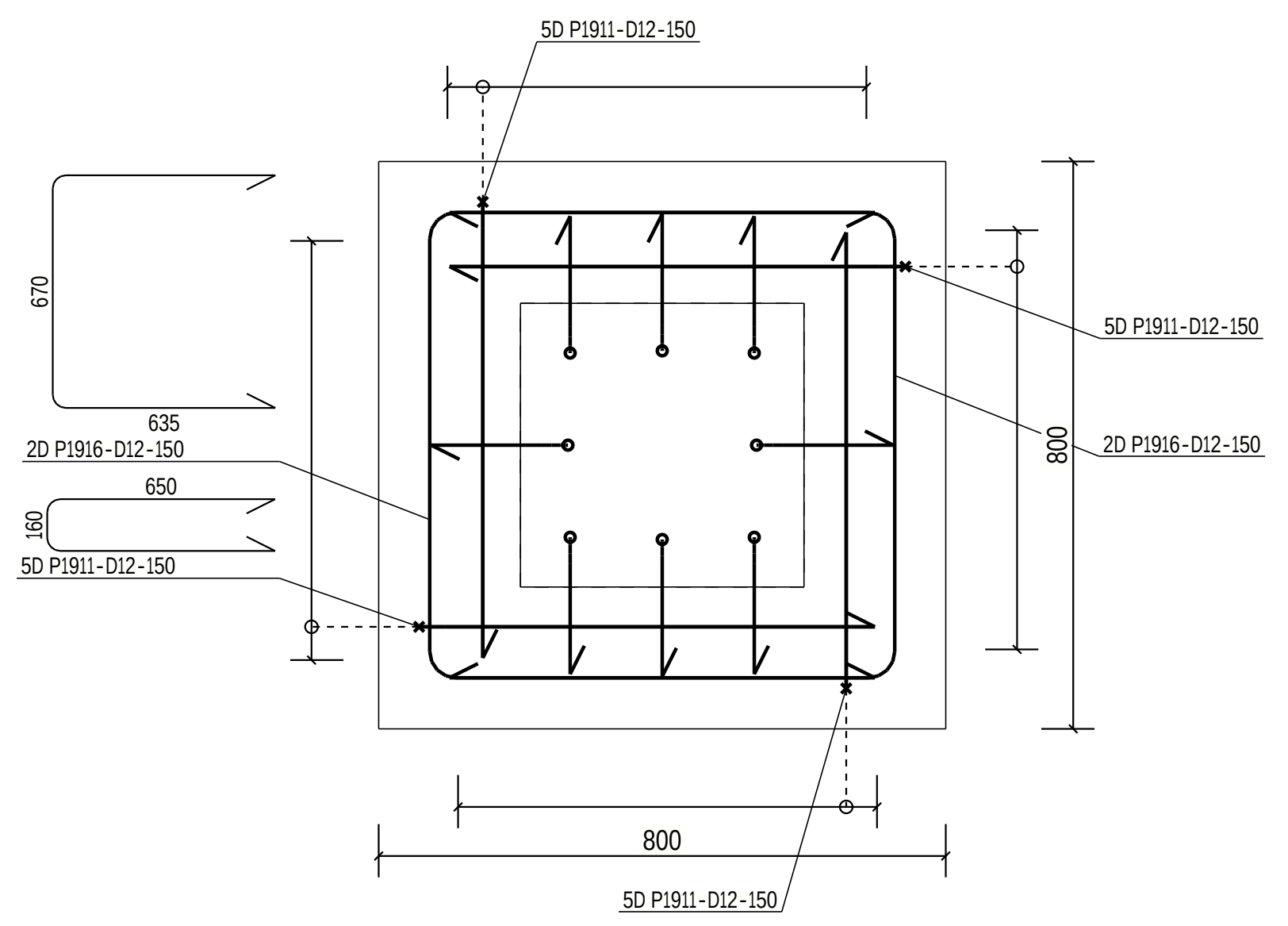
SECTION : 1 - 1
(SCALE 1:20)


2*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
3D P1915-D12
280
500
T.O.C. +913.725
2
2
P191
200
8*1B P1913-D12
F.G.L. +912.930 VARIES
ROAD
1150
5D P1911-D12-150
5D P1911-D12-150
3
3
3D P1911-D12-150
2D P1916-D12-150
B.O.F. +912.275
GROUT

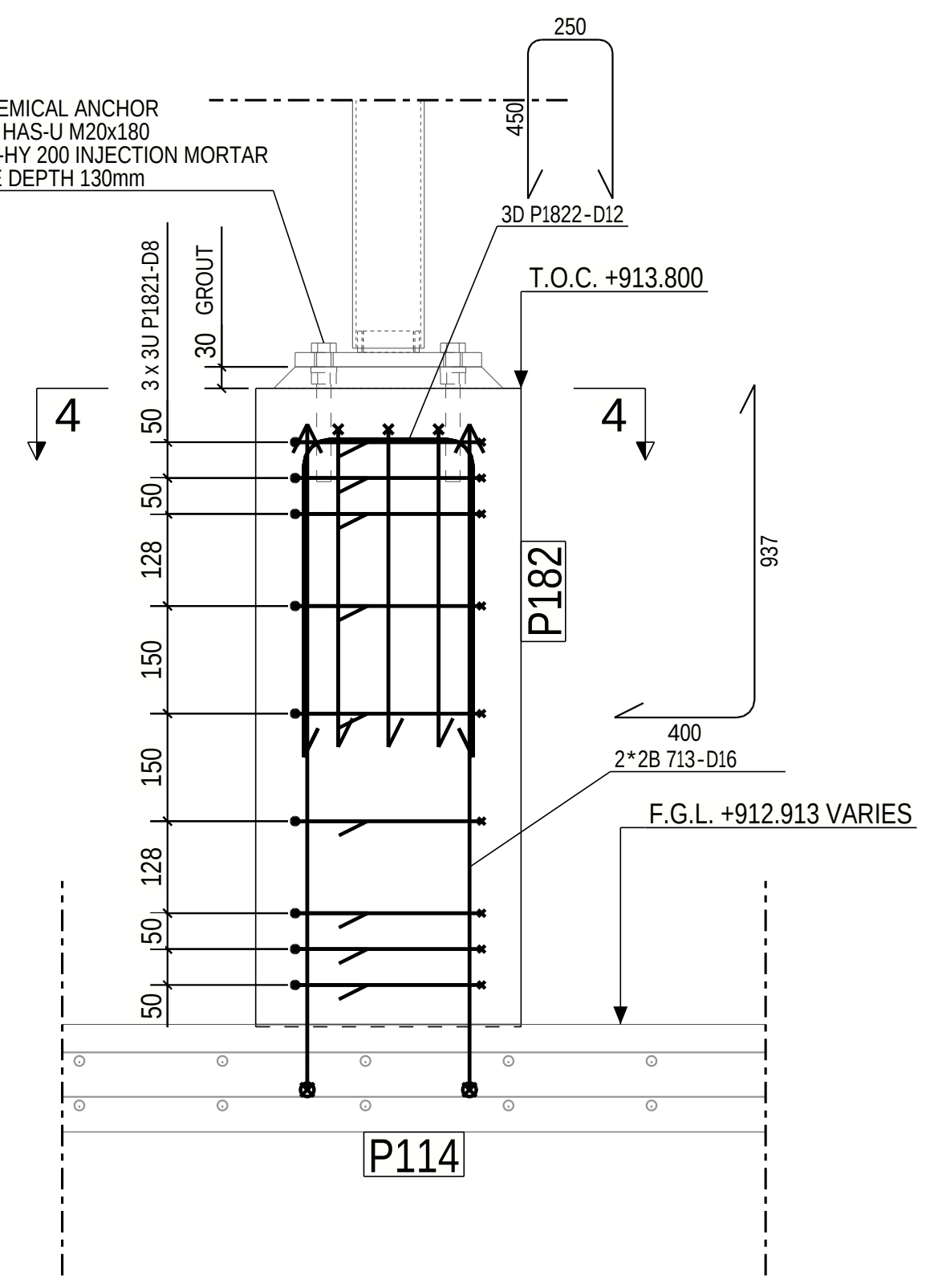
FOUNDATION P191 REINFORCEMENT
(SCALE 1:10)


3D P1915-D12
400
70 130 130 70
8B P1913-D12
106 300
3U P1912-D8
500
3D P1915-D12
6D P1915-D12 110 105 75 110

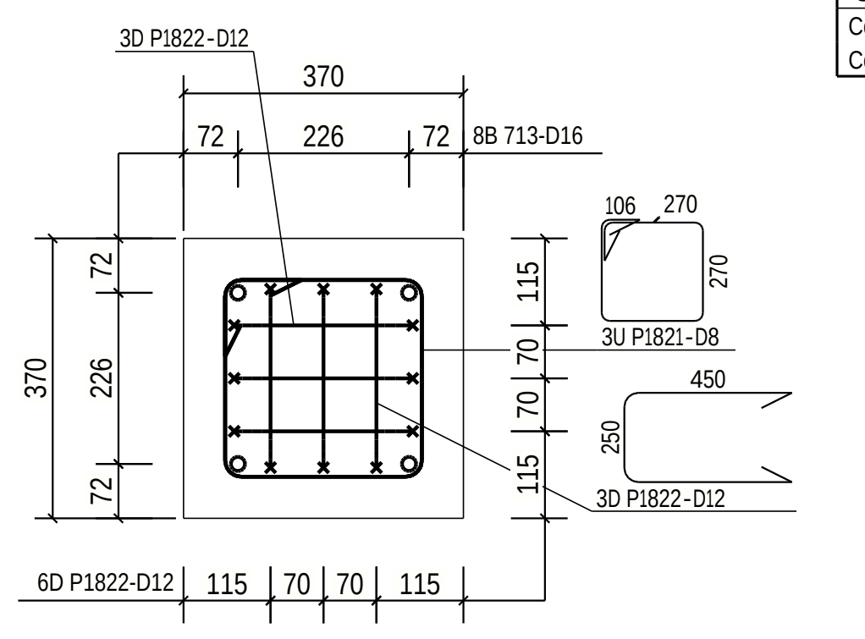
SECTION : 2 - 2
(SCALE 1:10)


5D P1911-D12-150
670
635
2D P1916-D12-150
650
5D P1911-D12-150
5D P1911-D12-150
2D P1916-D12-150
800
800
5D P1911-D12-150

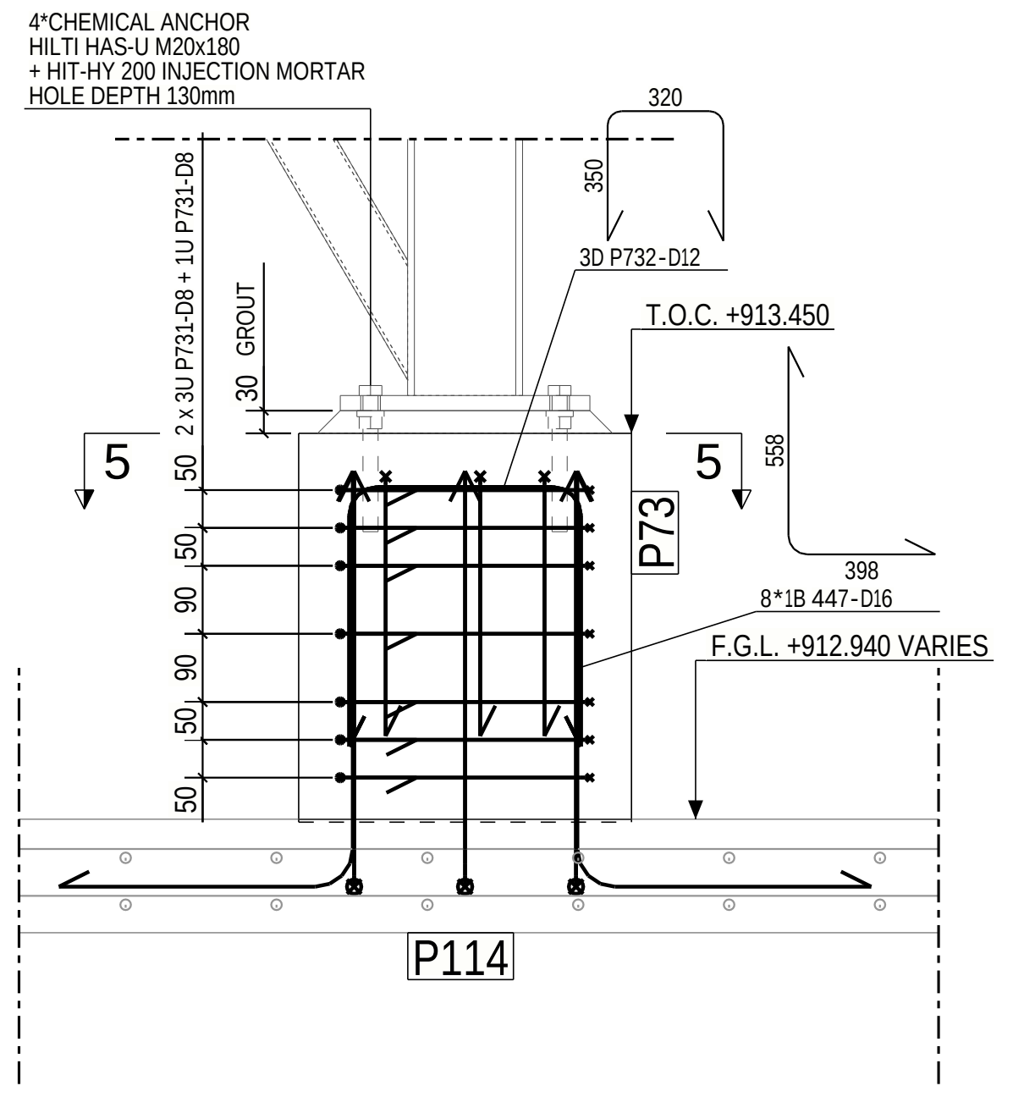
SECTION : 3 - 3
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
250
3D P1822-D12
T.O.C. +913.800
4
4
P182
400
2*2B 713-D16
F.G.L. +912.913 VARIES
P114

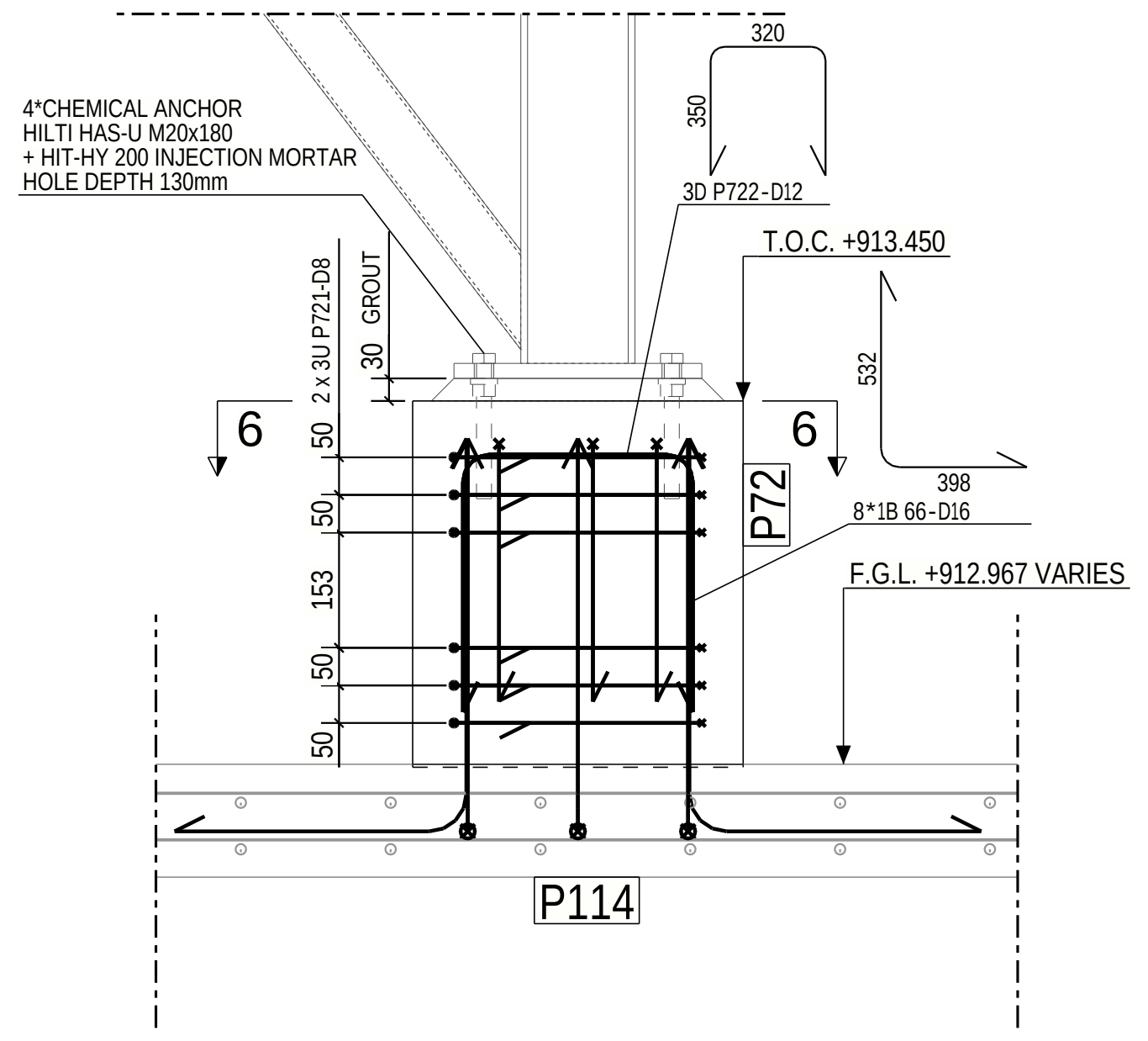
FOUNDATION P182 REINFORCEMENT
(SCALE 1:10)


3D P1822-D12
370
72 226 72
8B P1913-D16
106 270
3U P1821-D8
450
3D P1822-D12
6D P1822-D12 115 70 70 115

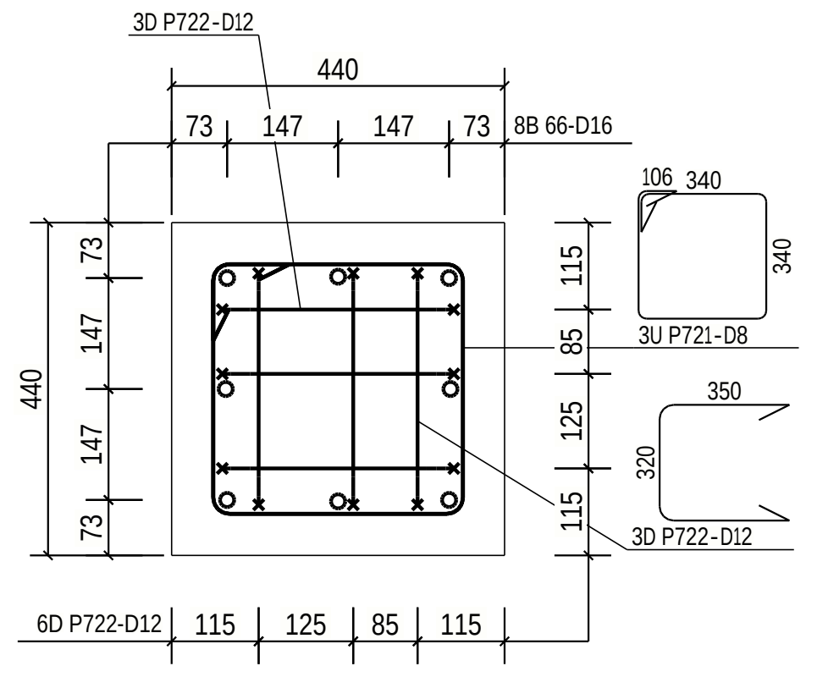
SECTION : 4 - 4
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
320
3D P732-D12
T.O.C. +913.450
5
5
P73
398
8*1B 447-D16
F.G.L. +912.940 VARIES
P114

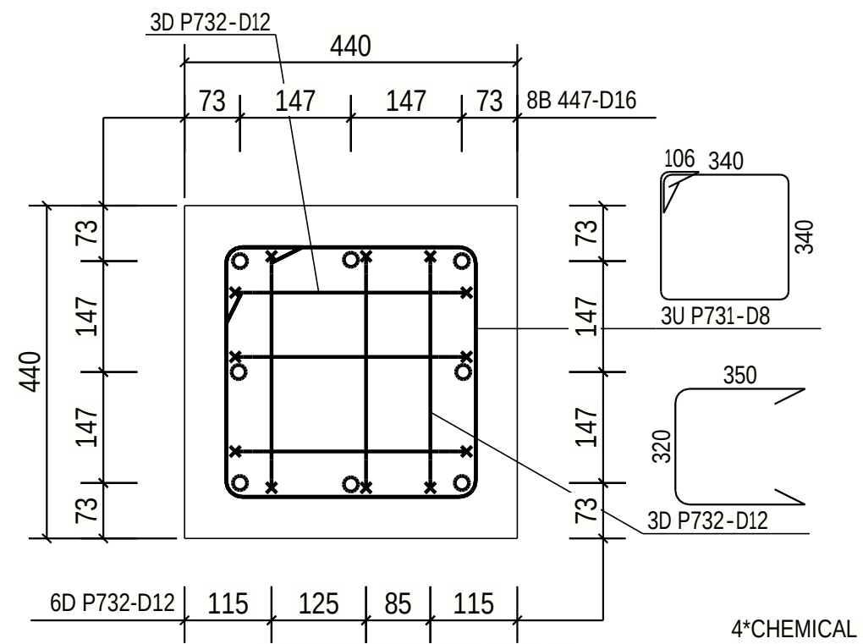
FOUNDATION P73 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
320
3D P722-D12
T.O.C. +913.450
6
6
P72
398
8*1B 66-D16
F.G.L. +912.967 VARIES
P114

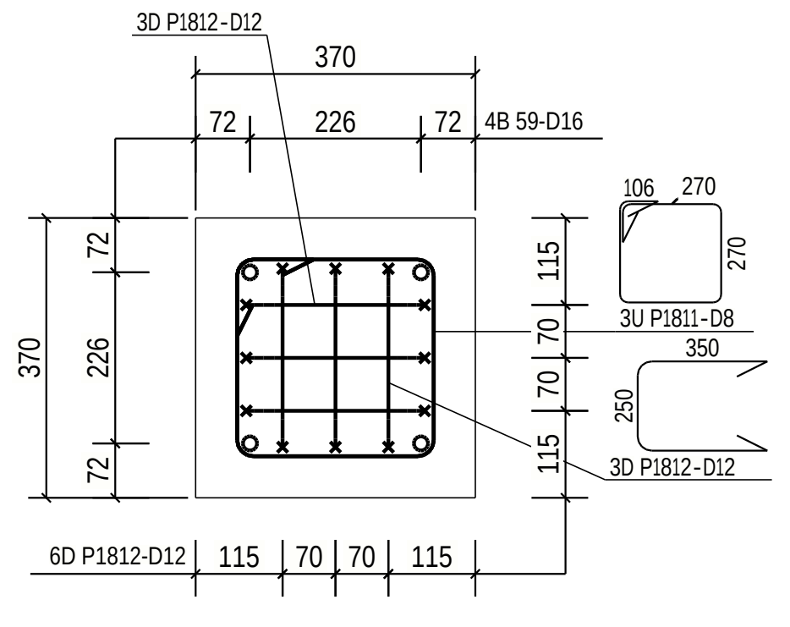
FOUNDATION P72 REINFORCEMENT
(SCALE 1:10)


3D P722-D12
440
73 147 147 73
8B 66-D16
106 340
3U P721-D8
350
3D P722-D12
6D P722-D12 115 125 85 115

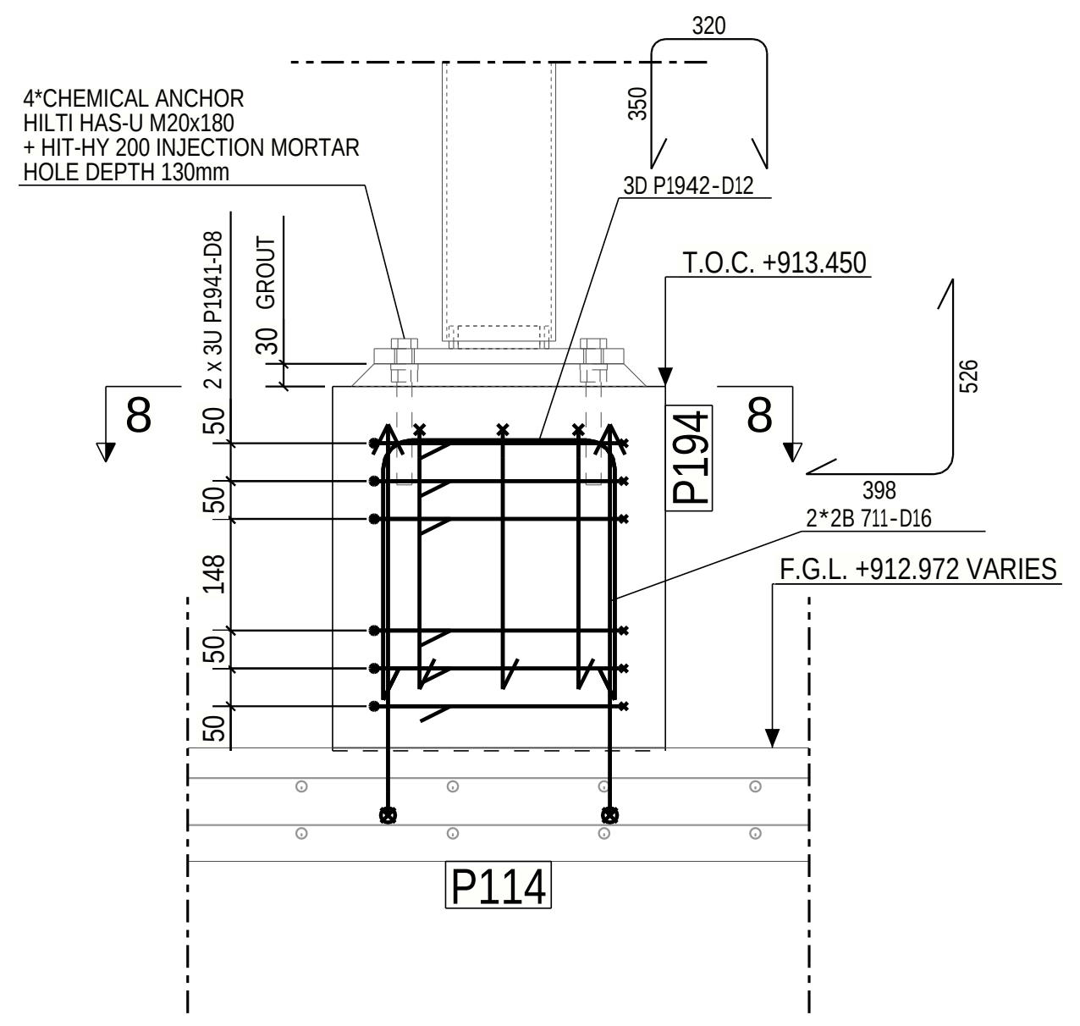
SECTION : 6 - 6
(SCALE 1:10)


3D P732-D12
440
73 147 147 73
8B 447-D16
106 340
3U P731-D8
350
3D P732-D12
6D P732-D12 115 125 85 115

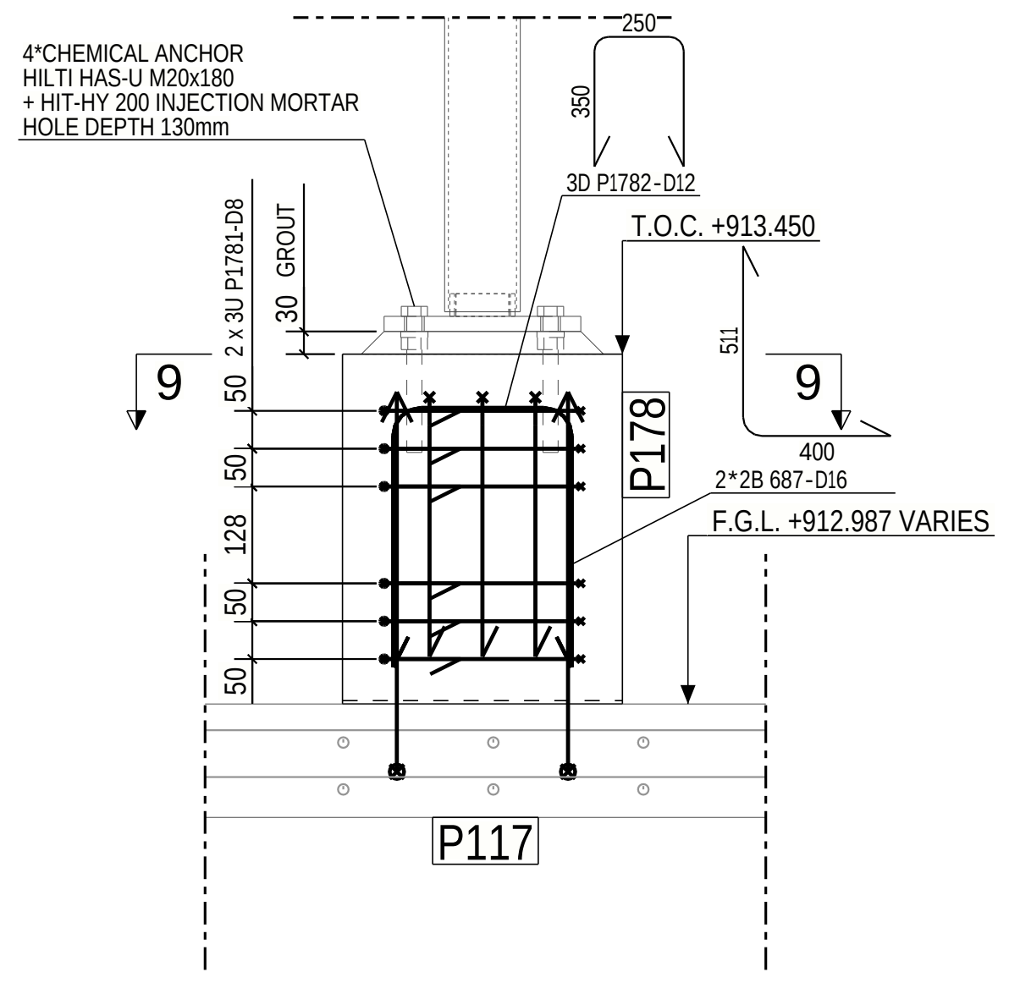
SECTION : 5 - 5
(SCALE 1:10)


3D P1812-D12
370
72 226 72
4B 59-D16
106 270
3U P1811-D8
350
3D P1812-D12
6D P1812-D12 115 70 70 115

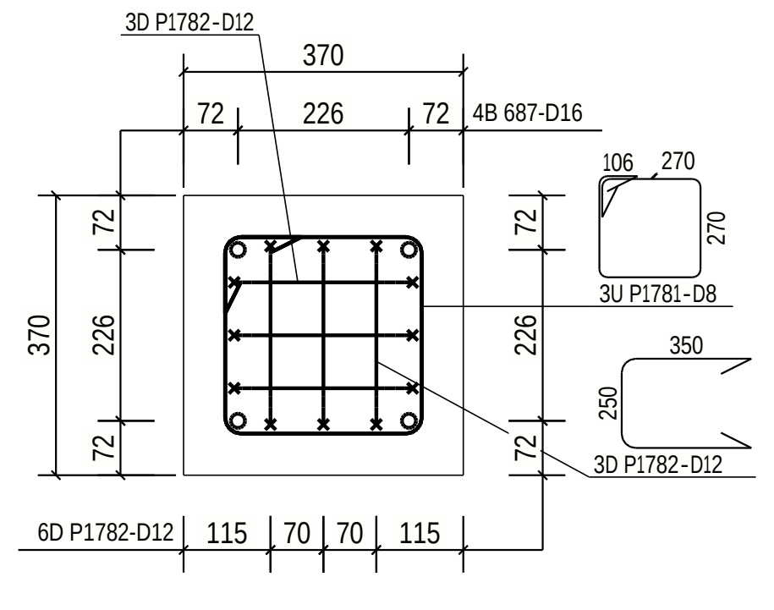
SECTION : 7 - 7
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
320
3D P1942-D12
T.O.C. +913.450
8
8
P194
398
2*2B 711-D16
F.G.L. +912.972 VARIES
P114

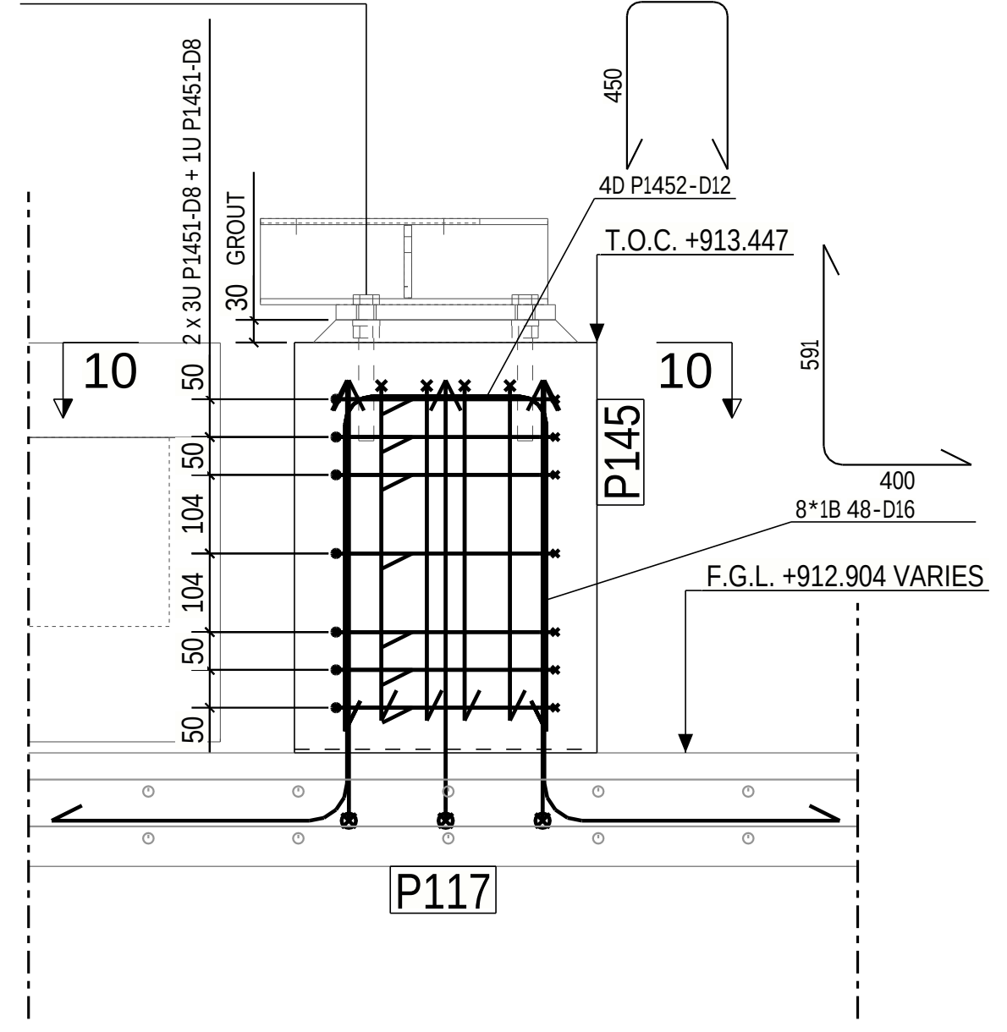
FOUNDATION P194 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
250
3D P1782-D12
T.O.C. +913.450
9
9
P178
400
2*2B 687-D16
F.G.L. +912.987 VARIES
P117

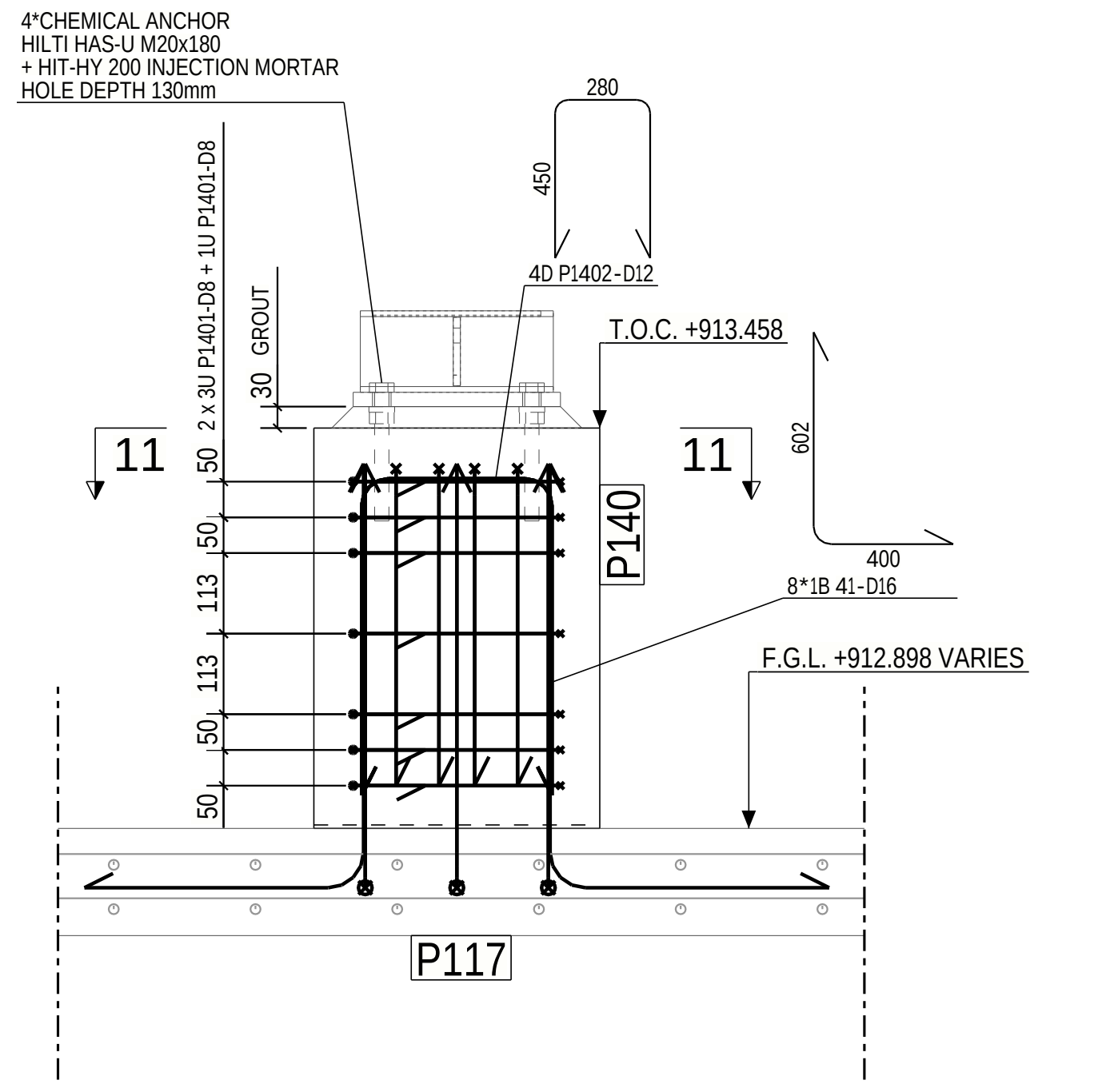
FOUNDATION P178 REINFORCEMENT
(SCALE 1:10)


3D P1782-D12
370
72 226 72
4B 687-D16
106 270
3U P1781-D8
350
3D P1782-D12
6D P1782-D12 115 70 70 115

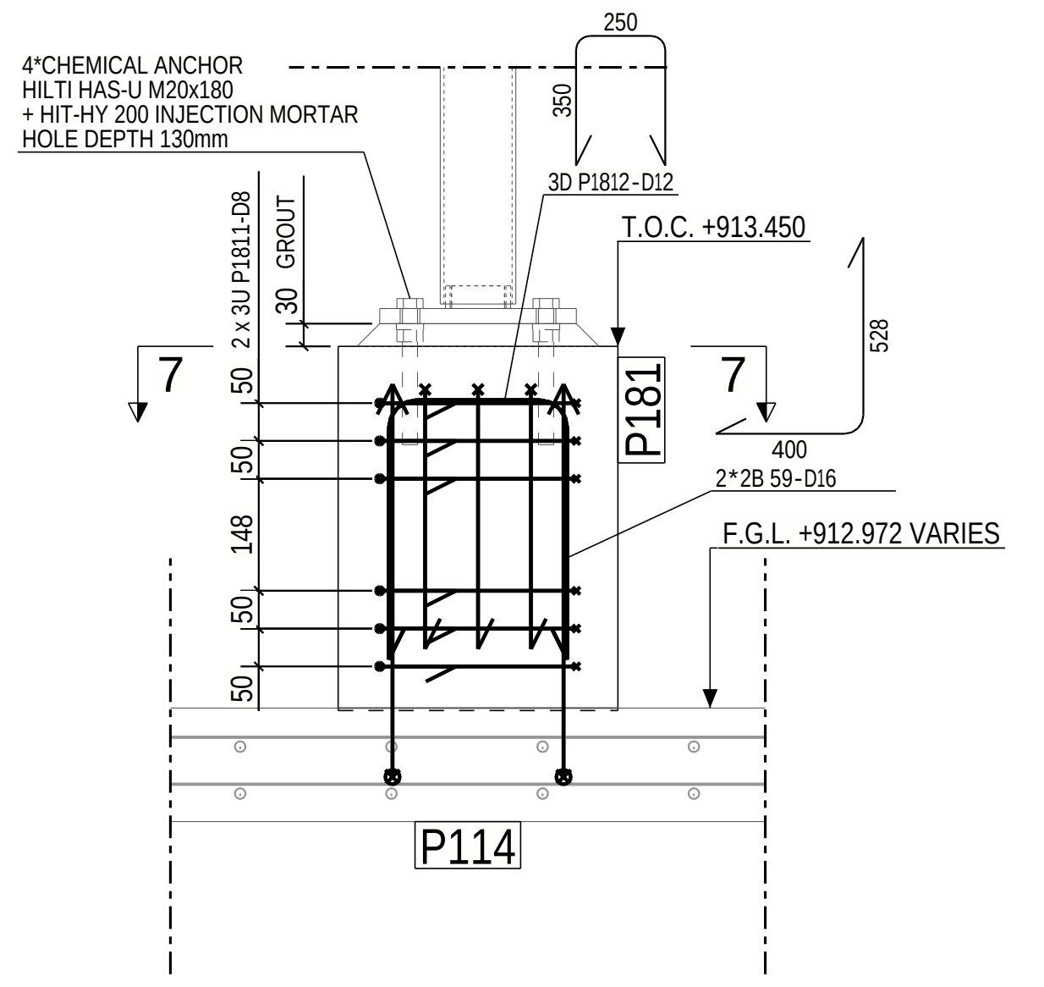
SECTION : 9 - 9
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
280
4D P1452-D12
T.O.C. +913.447
10
10
P145
400
8*1B 48-D16
F.G.L. +912.904 VARIES
P117

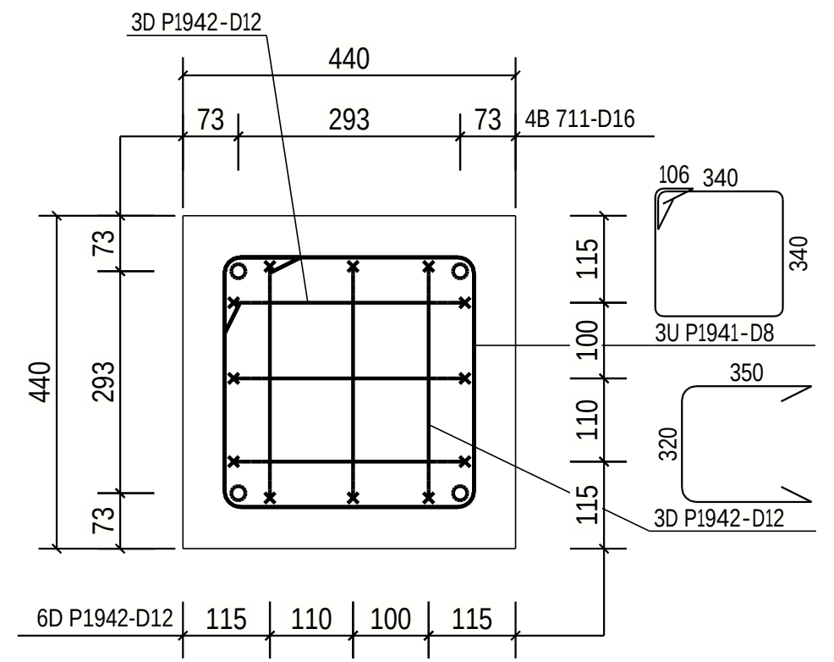
FOUNDATION P145 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
280
4D P1402-D12
T.O.C. +913.458
11
11
P140
400
8*1B 41-D16
F.G.L. +912.898 VARIES
P117

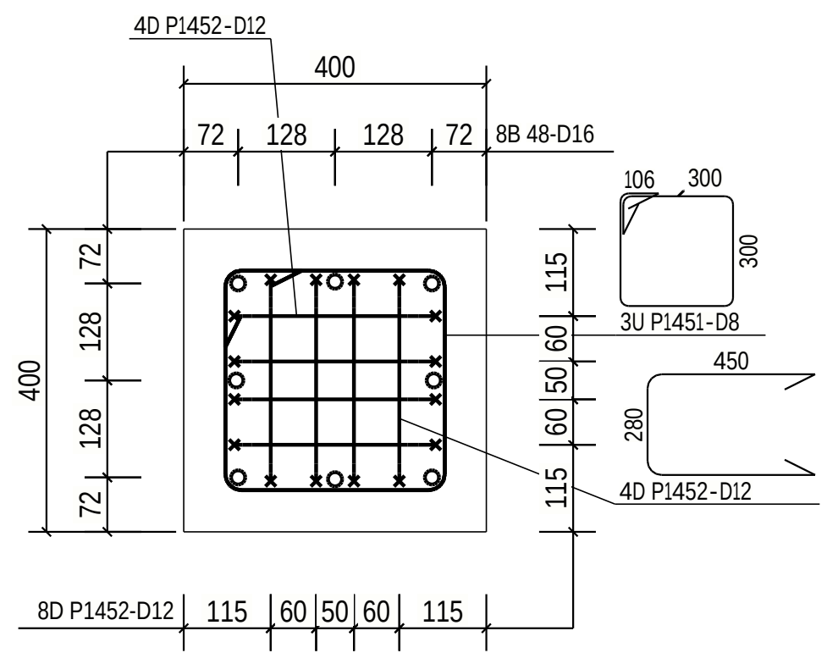
FOUNDATION P140 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
250
3D P1812-D12
T.O.C. +913.450
7
7
P181
400
2*2B 59-D16
F.G.L. +912.972 VARIES
P114

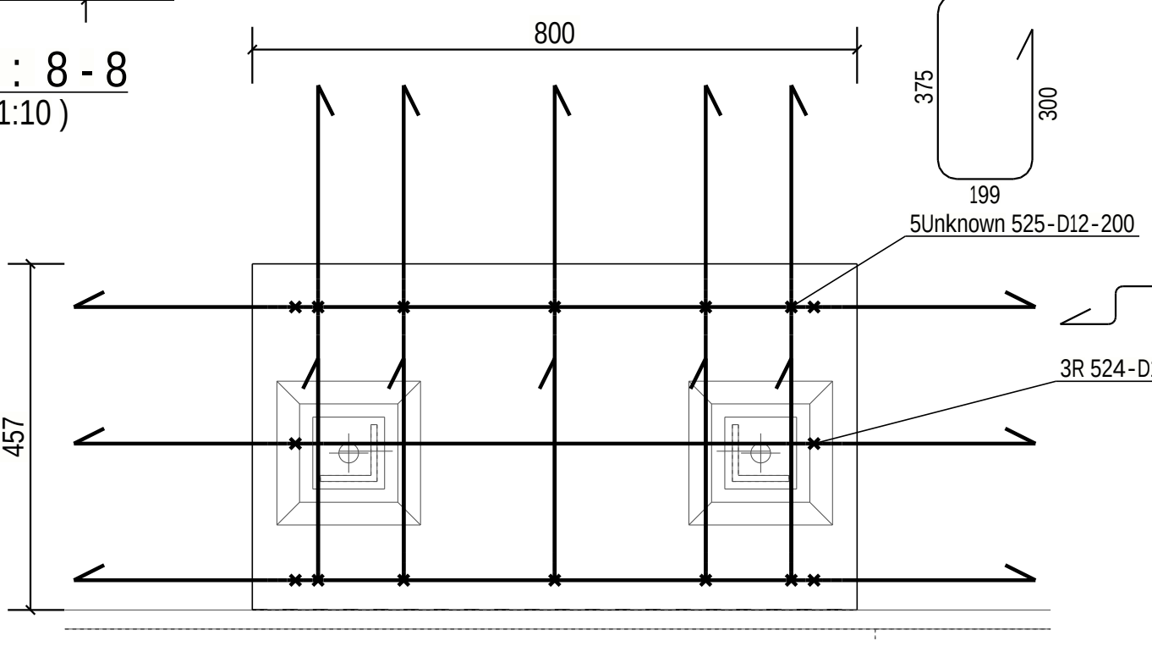
FOUNDATION P181 REINFORCEMENT
(SCALE 1:10)


3D P1942-D12
440
73 293 73
4B 711-D16
106 340
3U P1941-D8
350
3D P1942-D12
6D P1942-D12 115 110 100 115

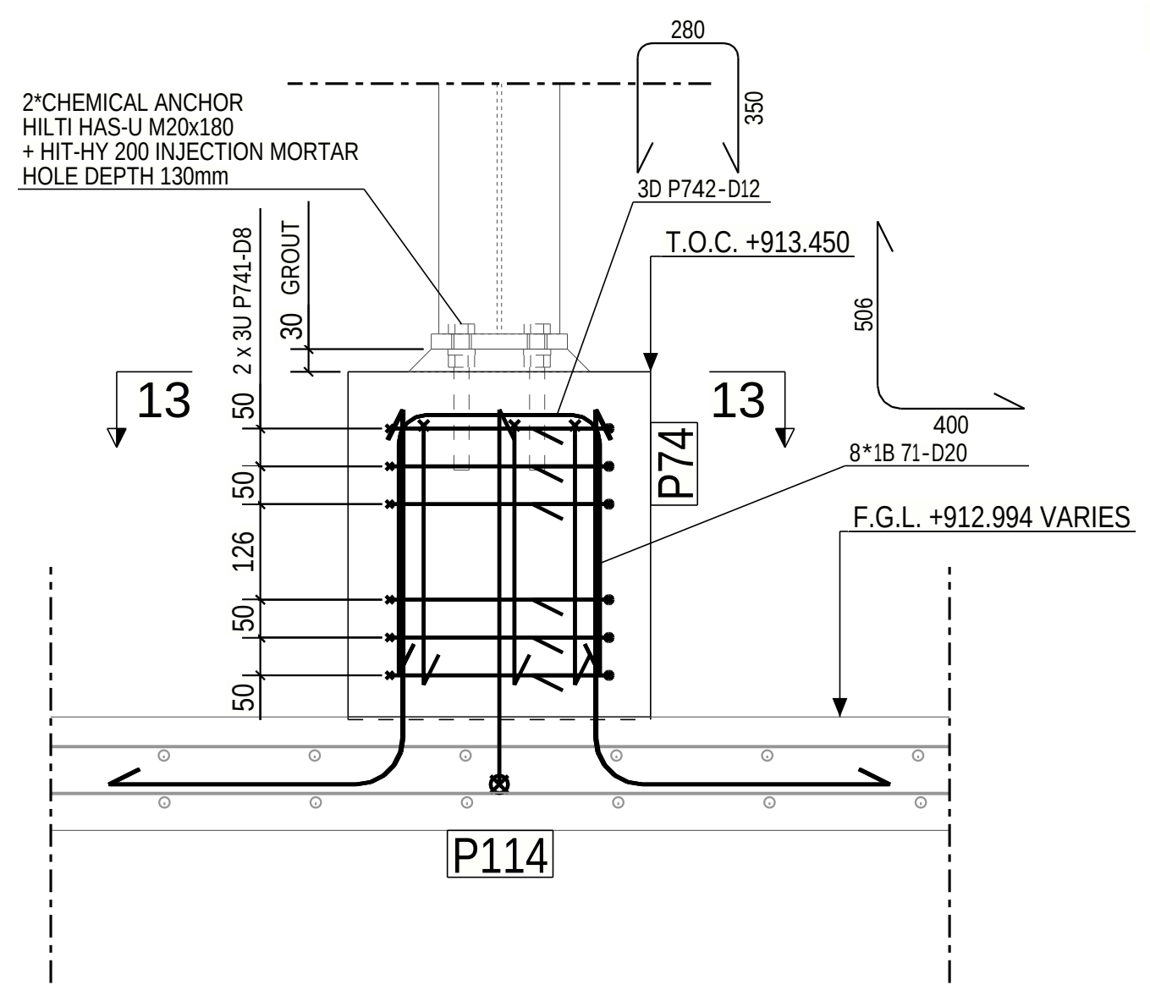
SECTION : 8 - 8
(SCALE 1:10)


4D P1452-D12
400
72 128 128 72
8B 48-D16
106 300
3U P1451-D8
450
4D P1452-D12
8D P1452-D12 115 60 50 60 115

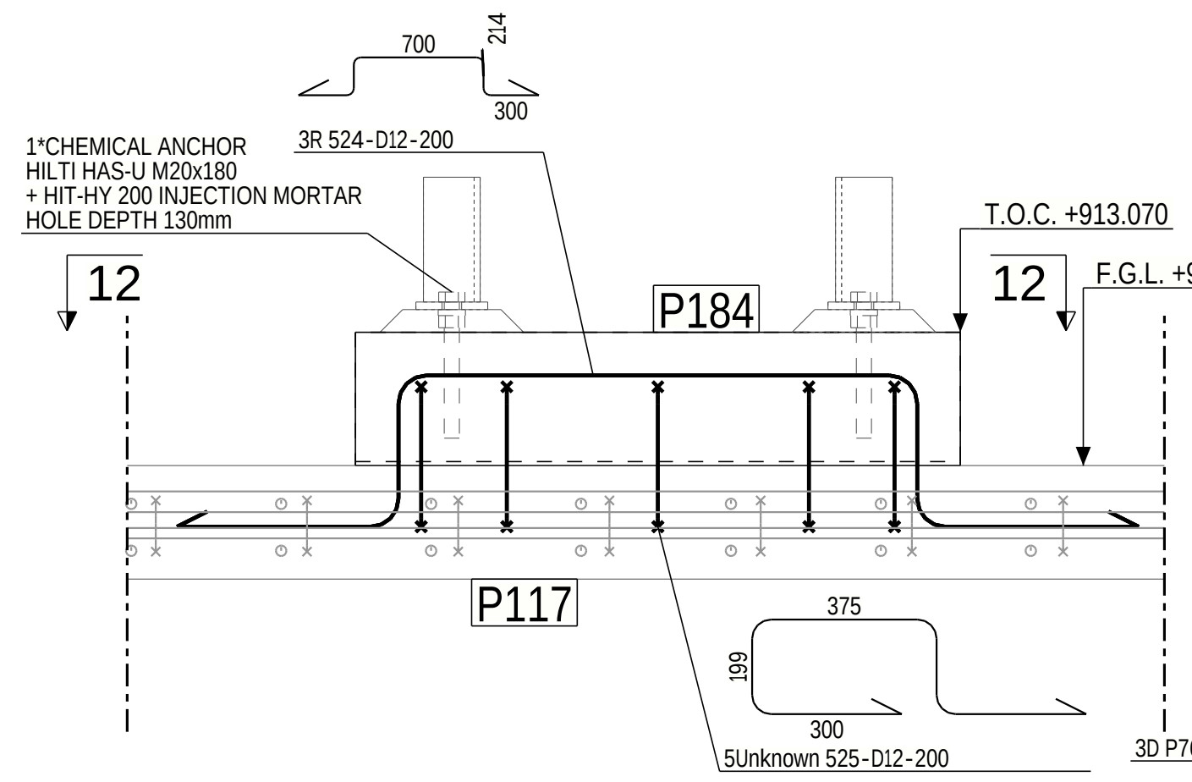
SECTION : 10 - 10
(SCALE 1:10)


800
199
5Unknown 525-D12-200
700
3R 524-D12-200
300
457

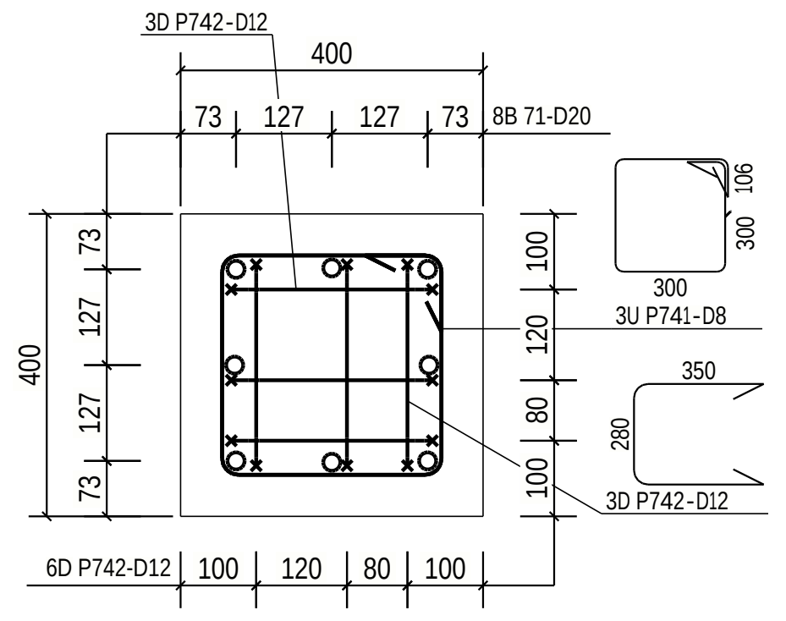
SECTION : 12 - 12
(SCALE 1:10)


2*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
280
3D P742-D12
T.O.C. +913.450
13
13
P74
400
8*1B 71-D20
F.G.L. +912.994 VARIES
P114

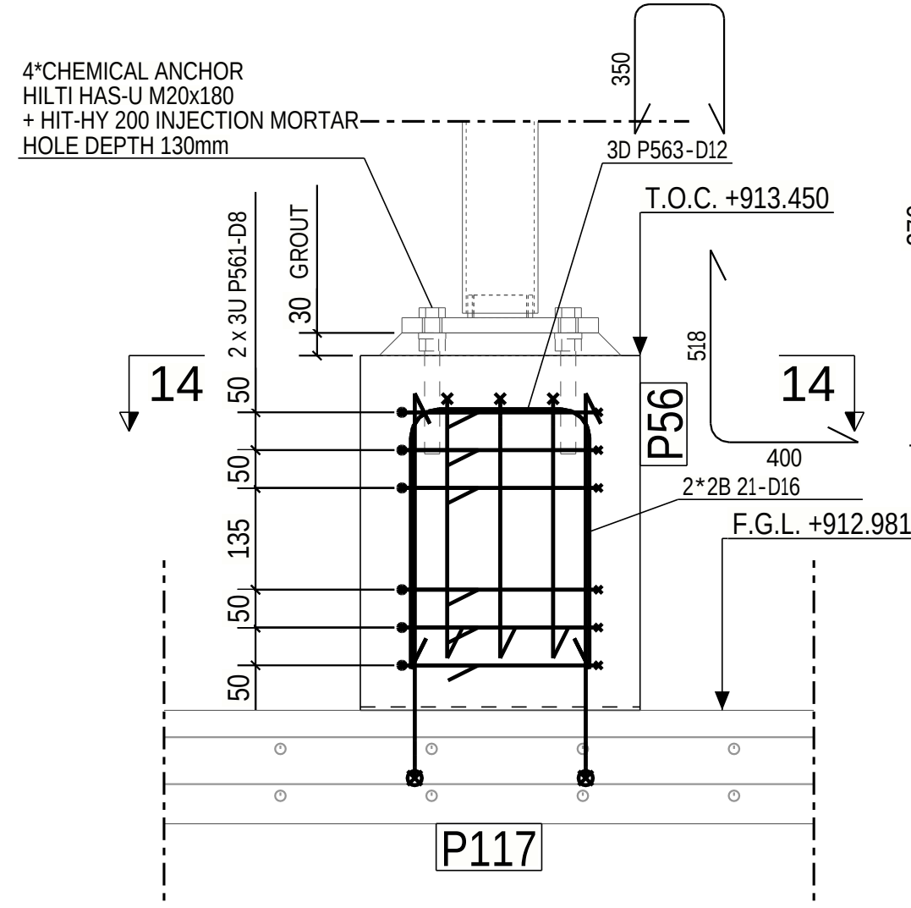
FOUNDATION P74 REINFORCEMENT
(SCALE 1:10)


1*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
700
3R 524-D12-200
300
T.O.C. +913.070
12
12
P184
F.G.L. +912.894 VARIES
P117
375
300
5Unknown 525-D12-200

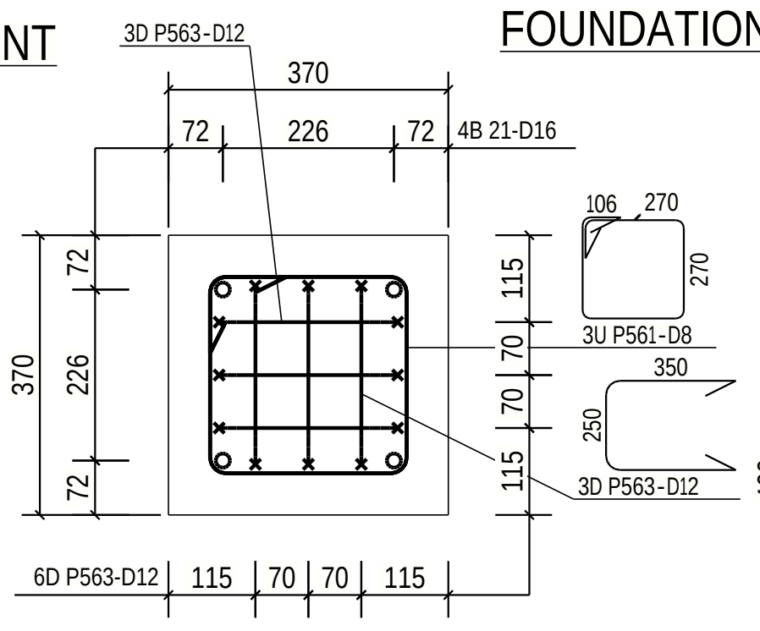
LADDER PLATFORM P184 REINFORCEMENT
(SCALE 1:10)


3D P742-D12
400
73 127 127 73
8B 71-D20
106 300
3U P741-D8
350
3D P742-D12
6D P742-D12 100 120 80 100

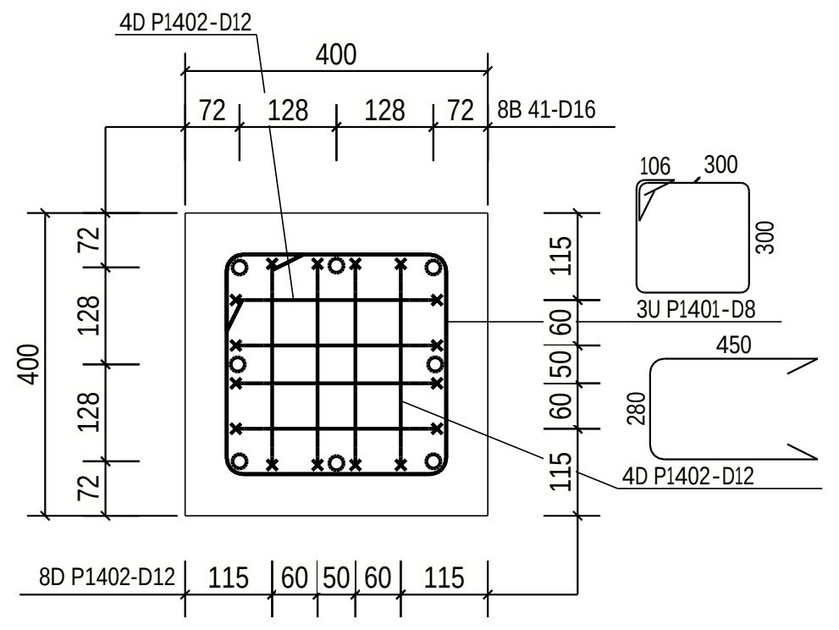
SECTION : 13 - 13
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
250
3D P563-D12
T.O.C. +913.450
14
14
P56
400
2*2B 21-D16
F.G.L. +912.981 VARIES
P117

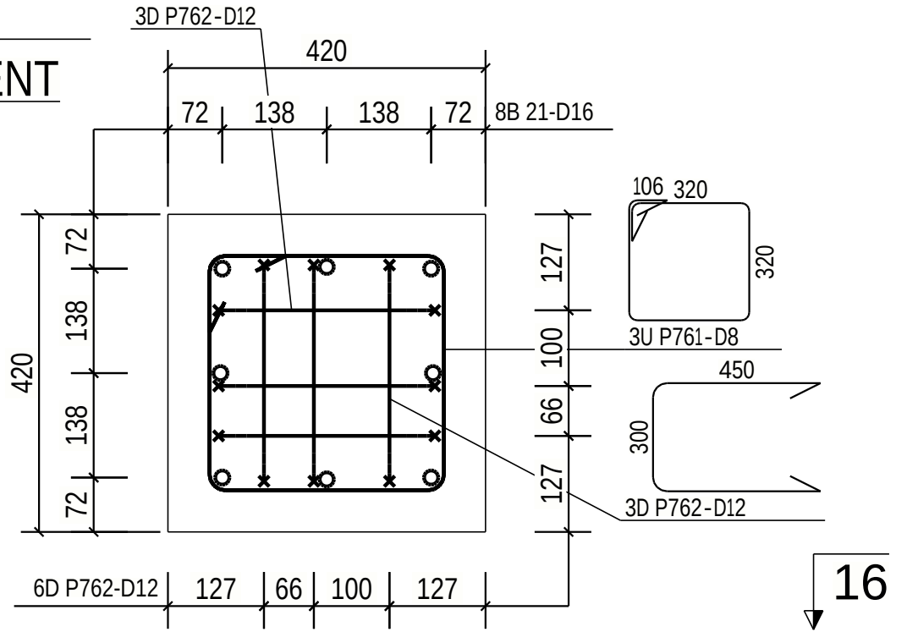
FOUNDATION P56 REINFORCEMENT
(SCALE 1:10)


3D P563-D12
370
72 226 72
4B 21-D16
106 270
3U P561-D8
350
3D P563-D12
6D P563-D12 115 70 70 115

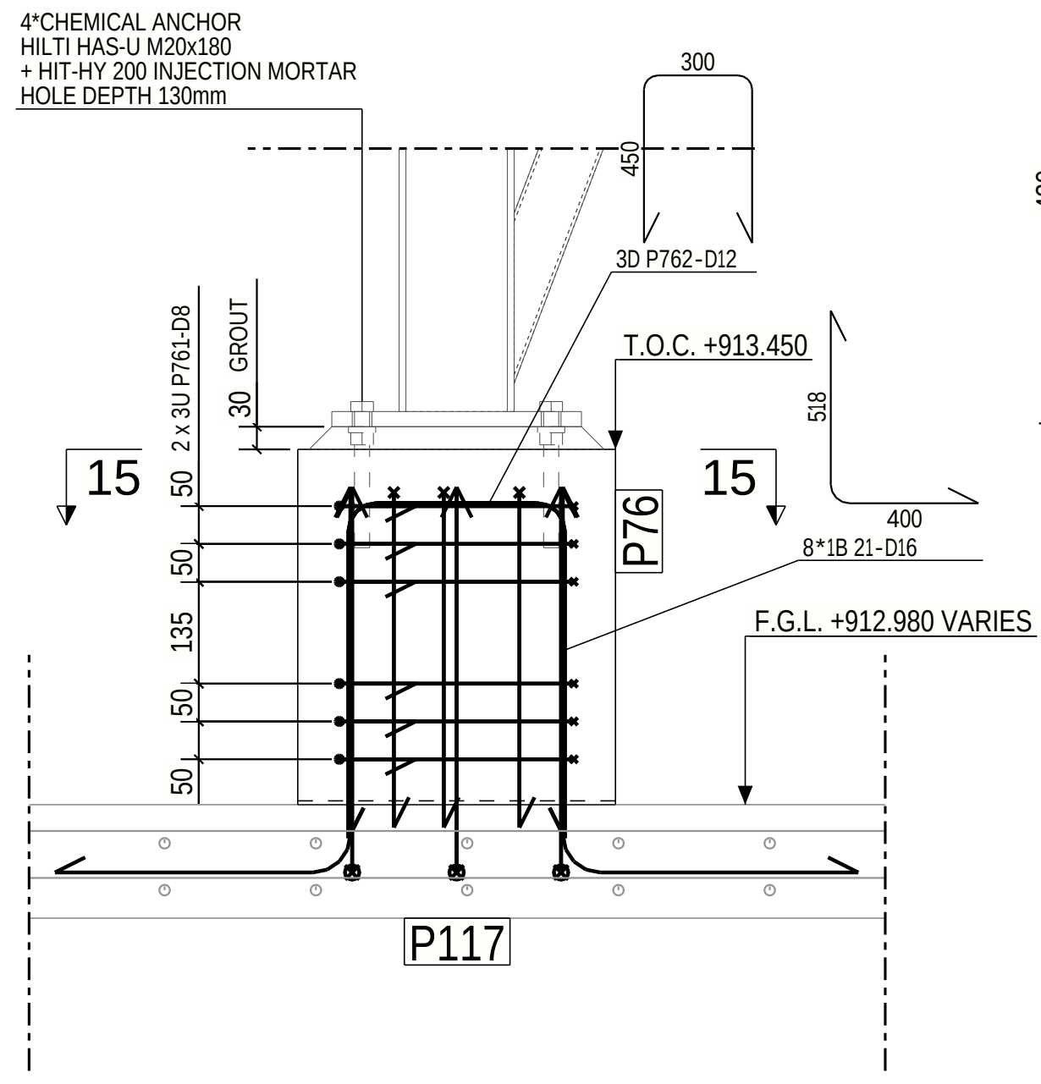
SECTION : 14 - 14
(SCALE 1:10)


4D P1402-D12
400
72 128 128 72
8B 41-D16
106 300
3U P1401-D8
450
4D P1402-D12
8D P1402-D12 115 60 50 60 115

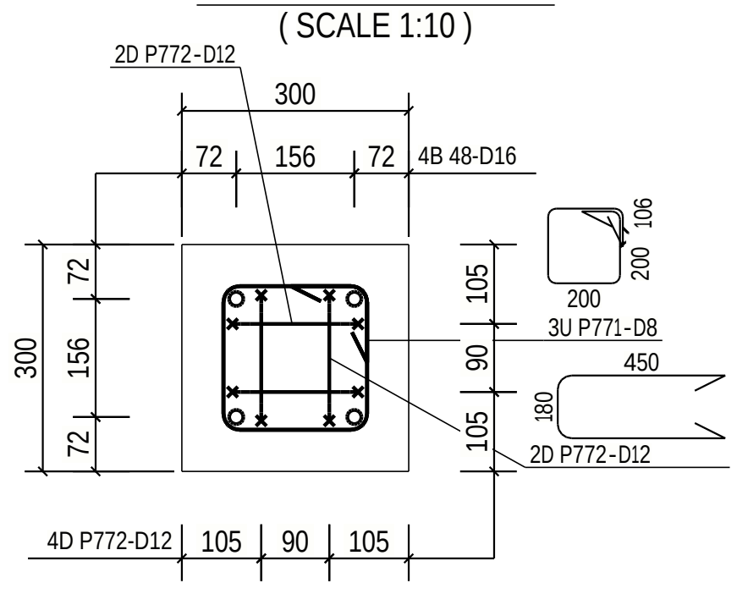
SECTION : 11 - 11
(SCALE 1:10)


3D P762-D12
420
72 138 138 72
8B 21-D16
106 320
3U P761-D8
300
3D P762-D12
6D P762-D12 127 66 100 127

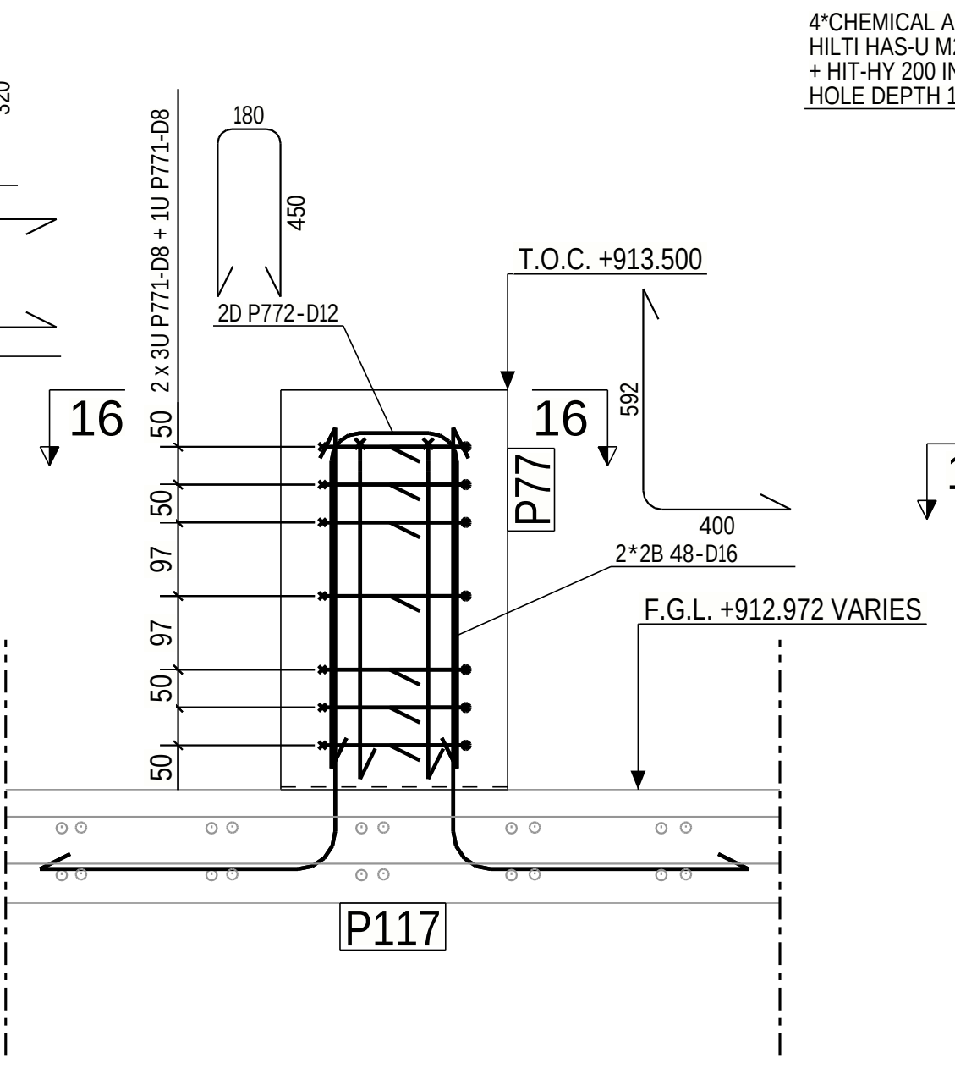
SECTION : 15 - 15
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
300
3D P762-D12
T.O.C. +913.450
15
15
P76
400
8*1B 21-D16
F.G.L. +912.980 VARIES
P117

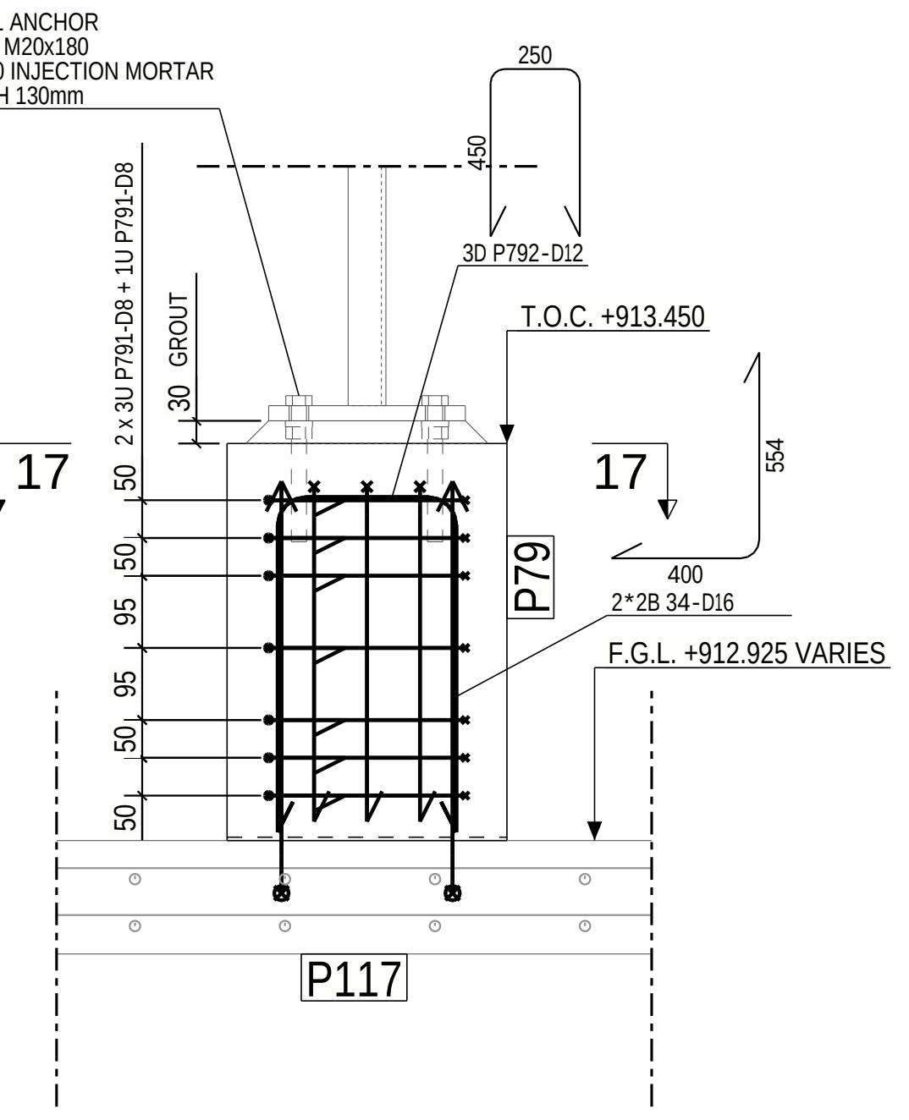
FOUNDATION P76 REINFORCEMENT
(SCALE 1:10)


2D P772-D12
300
72 156 72
4B 48-D16
200 106
3U P771-D8
450
2D P772-D12
4D P772-D12 105 90 105

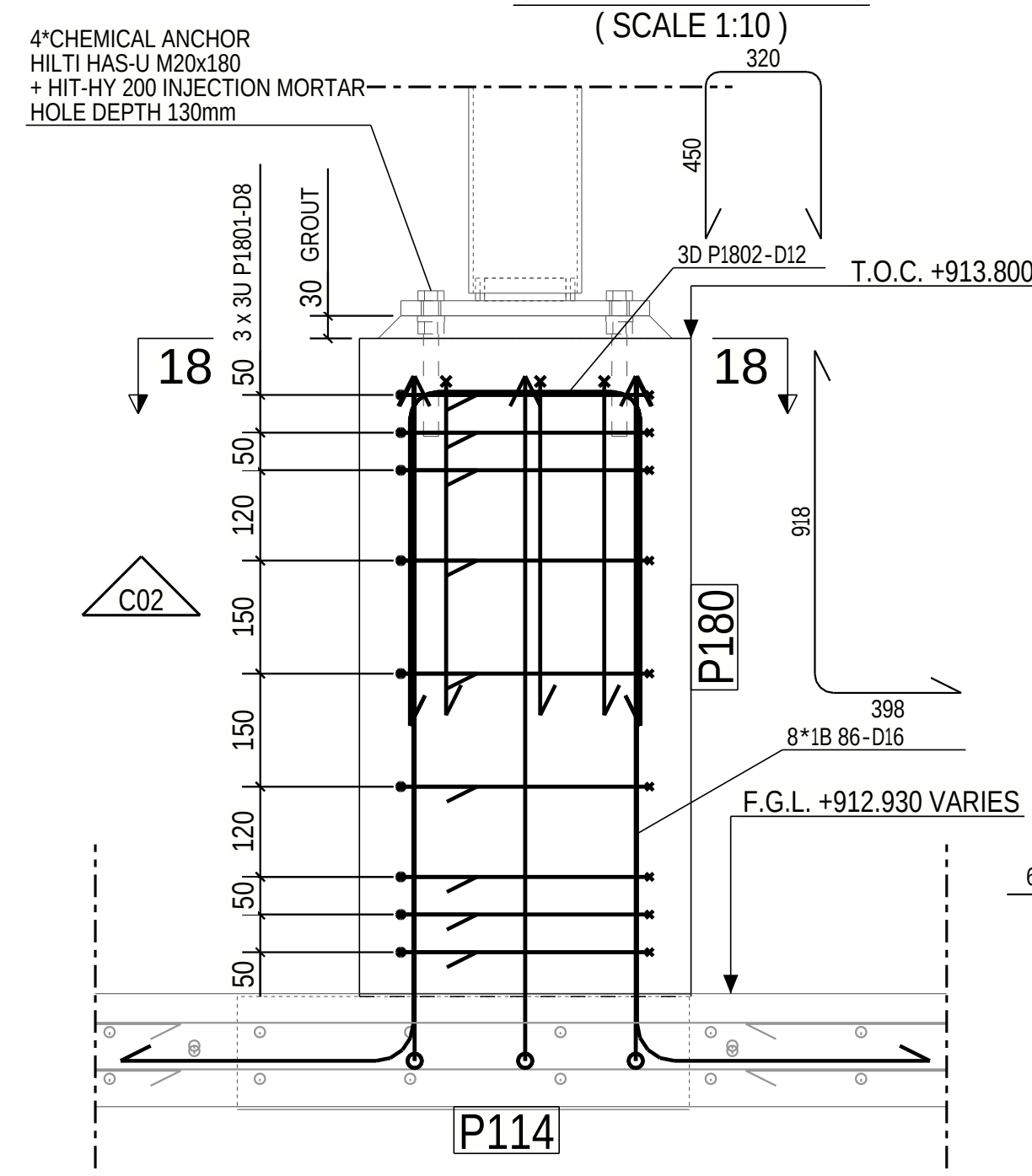
SECTION : 16 - 16
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
180
2D P772-D12
T.O.C. +913.500
16
16
P77
400
2*2B 48-D16
F.G.L. +912.972 VARIES
P117

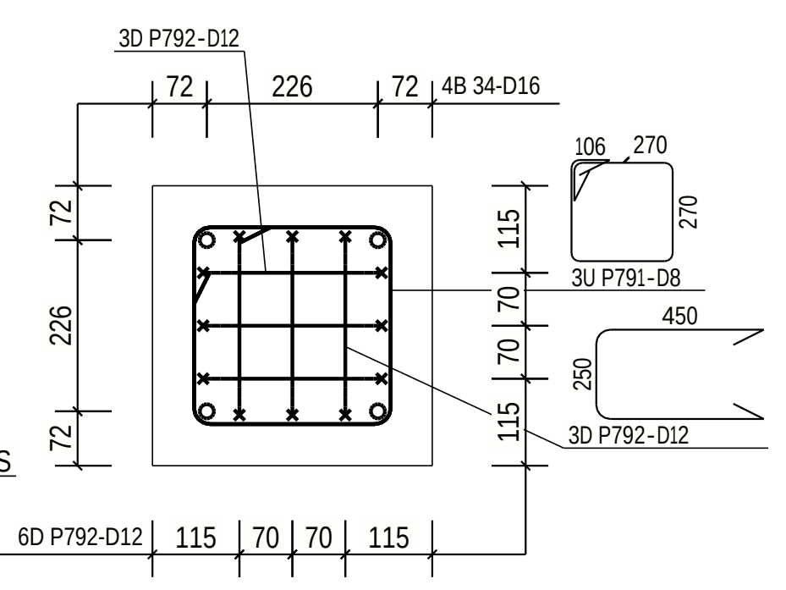
FOUNDATION P77 REINFORCEMENT
(SCALE 1:10)


250
T.O.C. +913.450
17
17
P79
400
2*2B 34-D16
F.G.L. +912.925 VARIES
P117

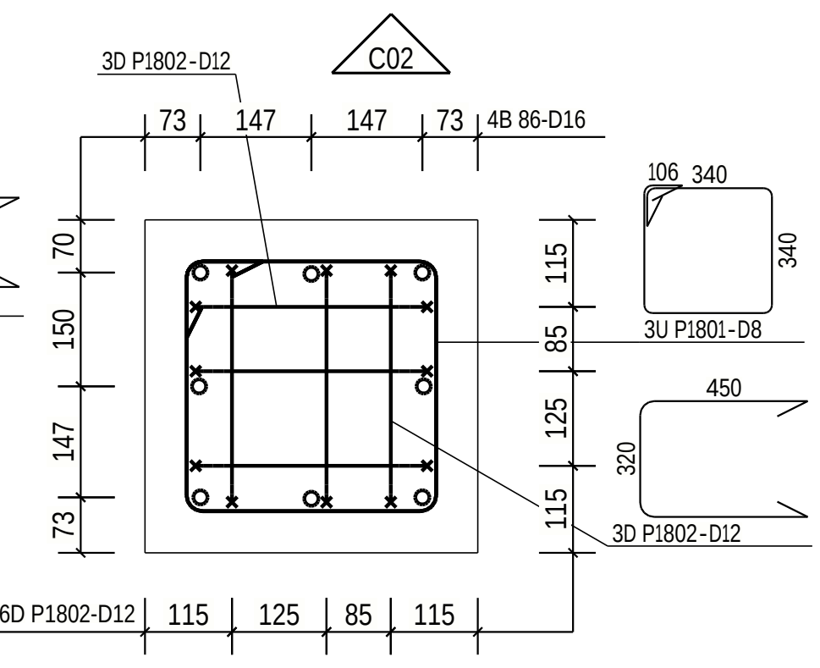
FOUNDATION P79 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR
HILTI HAS-U M20x180
+ HIT-HY 200 INJECTION MORTAR
HOLE DEPTH 130mm
320
3D P1802-D12
T.O.C. +913.800
18
18
C02
P180
398
8*1B 86-D16
F.G.L. +912.930 VARIES
P114

FOUNDATION P180 REINFORCEMENT
(SCALE 1:10)


3D P792-D12
226
72 72
4B 34-D16
106 270
3U P791-D8
450
3D P792-D12
6D P792-D12 115 70 70 115

SECTION : 17 - 17
(SCALE 1:10)


3D P1802-D12
C02
73 147 147 73
4B 86-D16
106 340
3U P1801-D8
450
3D P1802-D12
6D P1802-D12 115 125 85 115

SECTION : 18 - 18
(SCALE 1:10)

NOTE: LOCATIONS FOR STARTER BARS ARE SHOWN IN DRAWINGS:
MD-216-49W1-EG-CV-DED-1028
MD-216-49W1-EG-CV-DED-1031
AND NOT SHOWN IN REINFORCEMENT LIST

LOCATIONS FOR STRUCTURES IN GENERAL ARRANGEMENT DRAWINGS:
MD-216-49W1-EG-CV-GAD-1001
MD-216-49W1-EG-CV-GAD-1002

LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	50±10mm

REINFORCING BAR LIST — CAST UNIT CODE: P56 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P561	6	B500B	8	1190		2.8	106	270	270	270	270				32	
D	P563	6	B500B	12	890		4.7	350	250	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.6

REINFORCING BAR LIST — CAST UNIT CODE: P72 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P721	6	B500B	8	1470		3.5	106	340	340	340	340				32	
D	P722	6	B500B	12	960		5.1	350	320	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 8.6

REINFORCING BAR LIST — CAST UNIT CODE: P73 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P731	7	B500B	8	1470		4.1	106	340	340	340	340				32	
D	P732	6	B500B	12	960		5.1	350	320	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.2

REINFORCING BAR LIST — CAST UNIT CODE: P74 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P741	6	B500B	8	1310		3.1	106	300	300	300	300				32	
D	P742	6	B500B	12	920		4.9	350	280	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 8.0

REINFORCING BAR LIST — CAST UNIT CODE: P76 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P761	6	B500B	8	1390		3.3	106	320	320	320	320				32	
D	P762	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.4

REINFORCING BAR LIST — CAST UNIT CODE: P77 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P771	7	B500B	8	910		2.5	106	200	200	200	200				32	
D	P772	4	B500B	12	1020		3.6	450	180	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 6.2

REINFORCING BAR LIST — CAST UNIT CODE: P79 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P791	7	B500B	8	1190		3.3	106	270	270	270	270				32	
D	P792	6	B500B	12	1090		5.8	450	250	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.1

REINFORCING BAR LIST — CAST UNIT CODE: P140 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1401	7	B500B	8	1310		3.6	106	300	300	300	300				32	
D	P1402	8	B500B	12	1120		8.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.6

REINFORCING BAR LIST — CAST UNIT CODE: P145 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1451	7	B500B	8	1310		3.6	106	300	300	300	300				32	
D	P1452	8	B500B	12	1120		8.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.6

REINFORCING BAR LIST — CAST UNIT CODE: P153 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
A	P1531	9	B500B	12	1400		11.2	1400									
A	P1532	11	B500B	12	1200		11.7	1200									

REINFORCING BAR TOTAL WEIGHT [kg] 22.9

REINFORCING BAR LIST — CAST UNIT CODE: P178 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1781	6	B500B	8	1190		2.8	106	270	270	270	270				32	
D	P1782	6	B500B	12	890		4.7	350	250	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.6

REINFORCING BAR LIST — CAST UNIT CODE: P180 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1801	9	B500B	8	1470		5.2	106	340	340	340	340				32	
D	P1802	6	B500B	12	1160		6.2	450	320	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.4

REINFORCING BAR LIST — CAST UNIT CODE: P181 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1811	6	B500B	8	1190		2.8	106	270	270	270	270				32	
D	P1812	6	B500B	12	890		4.7	350	250	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.6

REINFORCING BAR LIST — CAST UNIT CODE: P182 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1821	9	B500B	8	1190		4.3	106	270	270	270	270				32	
D	P1822	6	B500B	12	1090		5.8	450	250	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 10.1

REINFORCING BAR LIST — CAST UNIT CODE: P184 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P191 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
D	P1911	20	B500B	12	1400		24.9	650	160	650						60	
U	P1912	11	B500B	8	1310		5.7	106	300	300	300	300				32	
B	P1913	8	B500B	12	1480		10.5	200	1310				90			60	
D	P1915	6	B500B	12	1220		6.5	500	280	500						60	
D	P1916	4	B500B	12	1880		6.7	635	670	635						60	

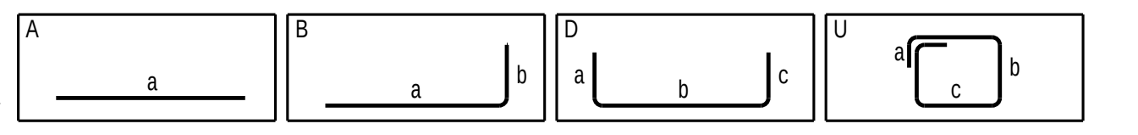
REINFORCING BAR TOTAL WEIGHT [kg] 54.3

REINFORCING BAR LIST — CAST UNIT CODE: P194 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1941	6	B500B	8	1470		3.5	106	340	340	340	340				32	
D	P1942	6	B500B	12	960		5.1	350	320	350						60	

REINFORCING BAR TOTAL WEIGHT [kg] 8.6

REINFORCEMENT BENDING SHAPES:


A
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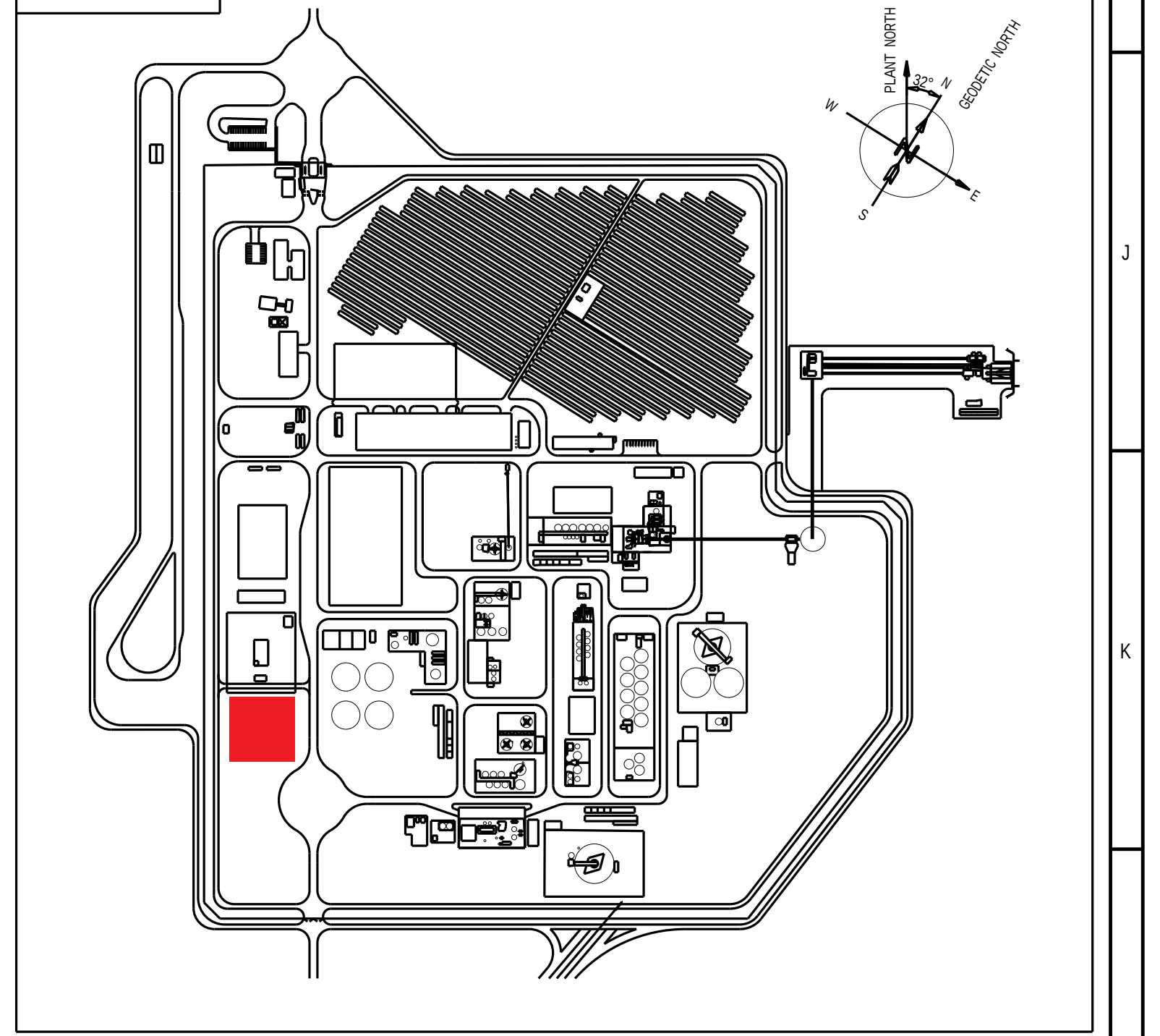
a
b
c

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1031	SLAB SECTOR GL D-G/1-4 DETAIL- AND REINFORCEMENT DRAWING		
MD-101-SMEM-EG-ST-DST-0001	CONCRETE STUCTURAL STANDARDS	MD-216-49W1-EG-CV-DED-1028	SLAB SECTOR GL G-J/1-5 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1003	1-10/A-E GENERAL ARRANGEMENT DRAWING	MD-216-1000-EG-ST-CRT-1001	DESIGN CRITERIA
MD-216-49W1-EG-CV-GAD-1001	1-8/E-J GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NanWu	LuiPiz	
C02	21.03.2022	FOUNDATION P180 ADDED	JonRen	MauNot	NanWu	LuiPiz	
C01	01.03.2022	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NanWu	LuiPiz	
A01	18.02.2022	ISSUED FOR APPROVAL	JonRen	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER
REVISION 3

OUTOTEC DRAWING NUMBER 900707_GAU01_AG11_CTB01_49038
DOCUMENT ID 0U100746295
REVISION 3

DRAWN	JonRen	18.02.2022
CHECKED	MauNot	18.02.2022
ENGINEER	NanWu	18.02.2022
APPROVED	LuiPiz	18.02.2022
EMPLOYER		
SCALE	1:15	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
SEC. PLATFORMS, STAIR AND PEDESTAL FOUNDATION 3
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER MD-216-49W1-EG-CV-DED-1038
REV D01